Q1 2021 (01 Sep Shift 2)

A student determined Young's Modulus of elasticity using the formula $Y=\frac{\mathrm{MgL}^3}{4 \mathrm{bd}^3 \delta}$. The value of g is taken to be $9.8 \mathrm{~m} / \mathrm{s}^2$, without any significant error, his observation are as following.

$Physical Quantity$	$Least count of the Equipment used for measurement$	$Observed value$
$Mass(M)$	1 g	2 kg
$Length of bar(L)$	1 mm	1 m
$Breadth of bar(b)$	0.1 mm	4 cm
$Thickness of bar(d)$	0.01 mm	0.4 cm
$Depression(8)$	0.01 mm	5 mm

Then the fractional error in the measurement of Y is:

(1) 0.0083

(2) 0.0155

(3) 0.155

(4) 0.083

Q2 2021 (31 Aug Shift 2)

Statement : **1**

If three forces $\vec{F}_1, \vec{F}_2$ and $\vec{F}_3$ are represented by three sides of a triangle and $\overrightarrow{\mathrm{F}}_1+\overrightarrow{\mathrm{F}}_2=-\overrightarrow{\mathrm{F}}_3$, then these three forces are concurrent forces and satisfy the condition for equilibrium.

Statement : II

A triangle made up of three forces $\vec{F}_1, \vec{F}_2$ and $\vec{F}_3$ as its sides taken in the same order, satisfy the condition for translatory equilibrium.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement-I is false but Statement-II is true

(2) Statement-I is true but Statement-II is false

(3) Both Statement-I and Statement-II are false

(4) Both Statement-I and Statement-II are true.

Q3 2021 (31 Aug Shift 2)

Statement I :

Two forces $(\overrightarrow{\mathrm{P}}+\overrightarrow{\mathrm{Q}})$ and $(\overrightarrow{\mathrm{P}}-\overrightarrow{\mathrm{Q}})$ where $\overrightarrow{\mathrm{P}} \perp \overrightarrow{\mathrm{Q}}$, when act at an angle θ_1 to each other, the magnitude of their resultant is $\sqrt{3\left(\mathrm{P}^2+\mathrm{Q}^2\right)}$, when they act at an angle θ_2, the magnitude of their resultant becomes $\sqrt{2\left(\mathrm{P}^2+\mathrm{Q}^2\right)}$. This is possible only when $\theta_1<\theta_2$.

Statement II :

In the situation given above.

$$\theta_1=60^{\circ} \text{ and } \theta_2=90^{\circ}$$

In the light of the above statements, choose the most appropriate answer from the options given below :-

(1) Statement-I is false but Statement-II is true

(2) Both Statement-I and Statement-II are true

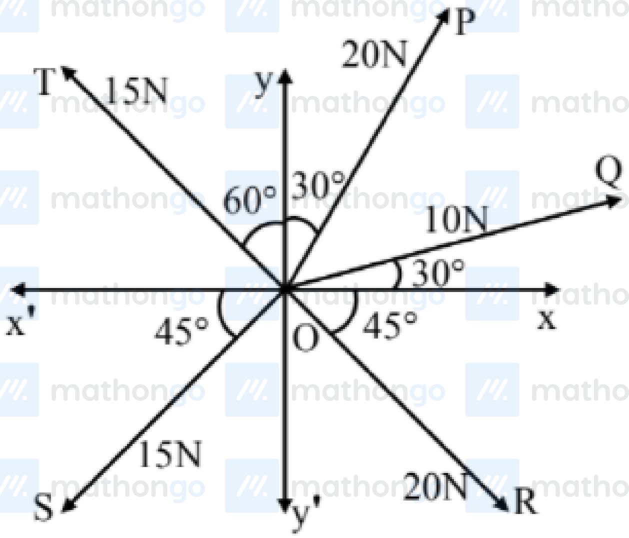
(3) Statement-I is true but Statement-II is false

(4) Both Statement-I and Statement-II are false.

Q4 2021 (27 Aug Shift 1)

The resultant of these forces $\overrightarrow{\mathrm{OP}}, \overrightarrow{\mathrm{OQ}}, \overrightarrow{\mathrm{OR}}, \overrightarrow{\mathrm{OS}}$ and $\overrightarrow{\mathrm{OT}}$ is approximately ..... N.

[Take $\sqrt{3}=1.7, \sqrt{2}=1.4$ Given $\hat{\mathrm{i}}$ and $\hat{\mathrm{j}}$ unit vectors along x, y axis]



(1) $9.25\hat{i}+5\hat{j}$

(2) $3\hat{i}+15\hat{j}$

(3) $2.5\hat{i}-14.5\hat{j}$

(4) $-1.5\hat{i}-15.5\hat{j}$

Q5 2021 (26 Aug Shift 2)

The acceleration due to gravity is found upto an accuracy of 4% on a planet. The energy supplied to a simple pendulum to known mass 'm' to undertake oscillations of time period T is being estimated. If time period is measured to an accuracy of 3%, the accuracy to which E is known as ....

Q6 2021 (26 Aug Shift 2)

If the length of the pendulum in pendulum clock increases by 0.1%, then the error in time per day is

(1) 86.4 s

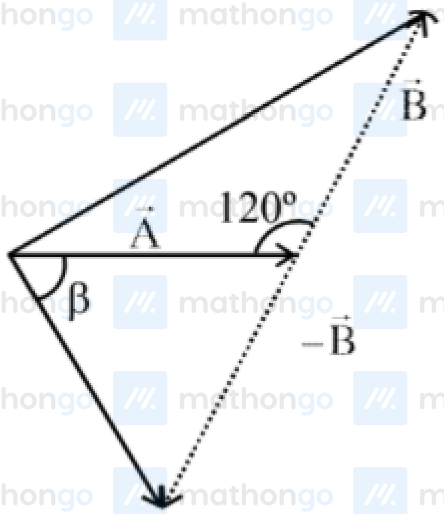
(2) 4.32 s

(3) 43.2 s

(4) 8.64 s

Q7 2021 (26 Aug Shift 2)

The angle between vector $(\vec{A})$ and $(\vec{A}-\vec{B})$ is :



(1) $\tan^{-1}\left(\frac{-\frac{B}{2}}{A-B\frac{\sqrt{3}}{2}}\right)$

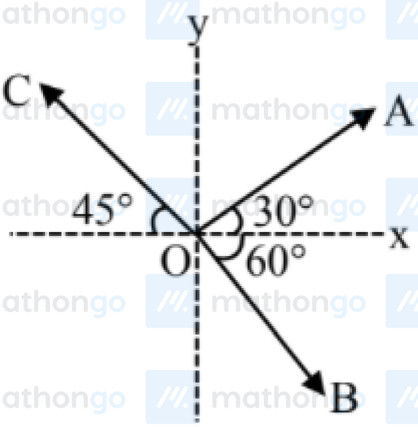
(2) $\tan^{-1}\left(\frac{A}{0.7\ B}\right)$

(3) $\tan^{-1}\left(\frac{\sqrt{3}\ B}{2\ A-B}\right)$

(4) $\tan^{-1}\left(\frac{B\cos\theta}{A-B\sin\theta}\right)$

Q8 2021 (26 Aug Shift 1)

The magnitude of vectors $\overrightarrow{OA}, \overrightarrow{OB}$ and $\overrightarrow{OC}$ in the given figure are equal. The direction of $\overrightarrow{OA}+\overrightarrow{OB}-\overrightarrow{OC}$ with x-axis will be :-



(1) $\tan^{-1}\frac{(1-\sqrt{3}-\sqrt{2})}{(1+\sqrt{3}+\sqrt{2})}$

(2) $\tan^{-1}\frac{(\sqrt{3}-1+\sqrt{2})}{(1+\sqrt{3}-\sqrt{2})}$

(3) $\tan^{-1}\frac{(\sqrt{3}-1+\sqrt{2})}{(1-\sqrt{3}+\sqrt{2})}$

(4) $\tan^{-1}\frac{(1+\sqrt{3}-\sqrt{2})}{(1-\sqrt{3}-\sqrt{2})}$

Answer Key

Q1 (2) **Q2** (4) **Q3** (2) **Q4** (1)

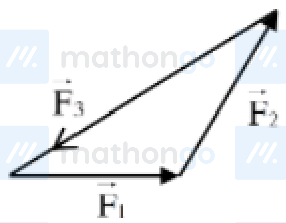
Q5 (14) **Q6** (3) **Q7** (3) **Q8** (1)

Q1 (2)

$$y=\frac{MgL^3}{4bd^3\delta}$$
$$\frac{\Delta y}{y}=\frac{\Delta M}{M}+\frac{3\Delta L}{L}+\frac{\Delta b}{b}+\frac{3\Delta d}{d}+\frac{\Delta\delta}{\delta}$$
$$\frac{\Delta y}{y}=\frac{10^{-3}}{2}+\frac{3\times10^{-3}}{1}+\frac{10^{-2}}{4}+\frac{3\times10^{-2}}{4}+\frac{10^{-2}}{5}$$
$$=10^{-3}[0.5+3+2.5+7.5+2]=0.0155$$

Option (2)

Q2 (4)



Here $\overrightarrow{\mathrm{F}}_1+\overrightarrow{\mathrm{F}}_2+\overrightarrow{\mathrm{F}}_3=0$

$$\overrightarrow{\mathrm{F}}_1+\overrightarrow{\mathrm{F}}_2=-\overrightarrow{\mathrm{F}}_3$$

Since $\vec{F}_{\text{nt}}=0$ (equilibrium)
Both statements correct

Q3 (2)

$\overrightarrow{\mathrm{A}}=\overrightarrow{\mathrm{P}}+\overrightarrow{\mathrm{Q}}$

$\overrightarrow{\mathrm{B}}=\overrightarrow{\mathrm{P}}-\overrightarrow{\mathrm{Q}}\quad\overrightarrow{\mathrm{P}}\perp\overrightarrow{\mathrm{Q}}$

$|\overrightarrow{\mathrm{A}}|=|\overrightarrow{\mathrm{B}}|=\sqrt{\mathrm{P}^2+\mathrm{Q}^2}$

$|\overrightarrow{\mathrm{A}}+\overrightarrow{\mathrm{B}}|=\sqrt{2\left(\mathrm{P}^2+\mathrm{Q}^2\right)(1+\cos\theta)}$

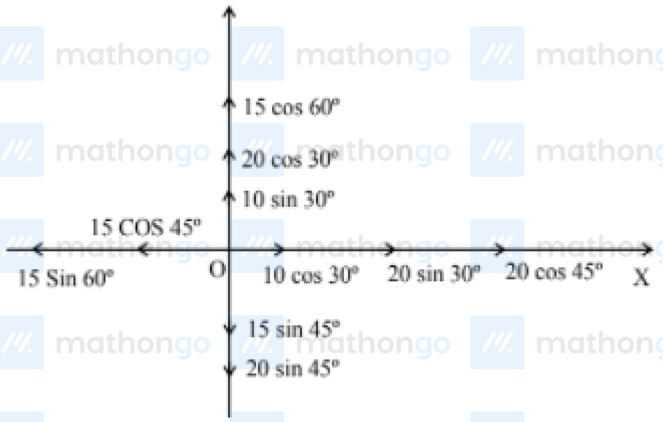
For $|\overrightarrow{\mathrm{A}}+\overrightarrow{\mathrm{B}}|=\sqrt{3\left(\mathrm{P}^2+\mathrm{Q}^2\right)}$

$\theta_1 = 60^\circ$

For $|\overrightarrow{\mathrm{A}} + \overrightarrow{\mathrm{B}}| = \sqrt{2\left(\mathrm{P}^2 + \mathrm{Q}^2\right)}$

$\theta_2 = 90^\circ$

Q4 (1)



$\overrightarrow{\mathrm{F}}_{\mathrm{x}} = \left(10 \times \frac{\sqrt{3}}{2} + 20\left(\frac{1}{2}\right) + 20\left(\frac{1}{\sqrt{2}}\right) - 15\left(\frac{1}{\sqrt{2}}\right) - 15\left(\frac{\sqrt{3}}{2}\right)\right)\hat{\mathrm{i}}$

$= 9.25\hat{\mathrm{i}}$

$\overrightarrow{\mathrm{F}}_{\mathrm{y}} = \left(15\left(\frac{1}{2}\right) + 20\left(\frac{\sqrt{3}}{2}\right) + 10\left(\frac{1}{2}\right) - 15\left(\frac{1}{\sqrt{2}}\right) - 20\left(\frac{1}{\sqrt{2}}\right)\right)\hat{\mathrm{j}}$

$= 5\hat{\mathrm{j}}$

Q5 (14)

$\mathrm{T} = 2\pi\sqrt{\frac{\ell}{\mathrm{g}}} \Rightarrow \ell = \frac{\mathrm{T}^2\,\mathrm{g}}{4\pi^2}$

$\mathrm{E} = \mathrm{mg}\ell\frac{\theta^2}{2} = \mathrm{mg}^2\,\frac{\mathrm{T}^2\theta^2}{8\pi^2}$

$\frac{\mathrm{dE}}{\mathrm{E}} = 2\left(\frac{\mathrm{dg}}{\mathrm{g}} + \frac{\mathrm{dT}}{\mathrm{T}}\right)$

$= (4+3) = 14\%$

Q6 (3)

$$T = 2\pi\sqrt{\frac{\ell}{g}}$$

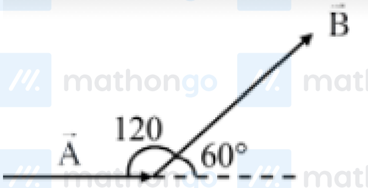
$$\frac{\Delta T}{T} = \frac{1}{2}\frac{\Delta \ell}{\ell}$$

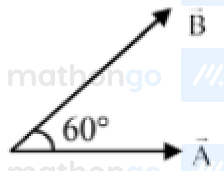
$$\Delta T = \frac{1}{2} \times \frac{0.1}{100} \times 24 \times 3600$$

$$\Delta T = 43.2$$

Ans. (3)

Q7 (3)





Angle between $\overrightarrow{A}$ and $\overrightarrow{B}$, $\theta = 60^\circ$

Angle betwenn $\overrightarrow{A}$ and $\overrightarrow{A} - \overrightarrow{B}$

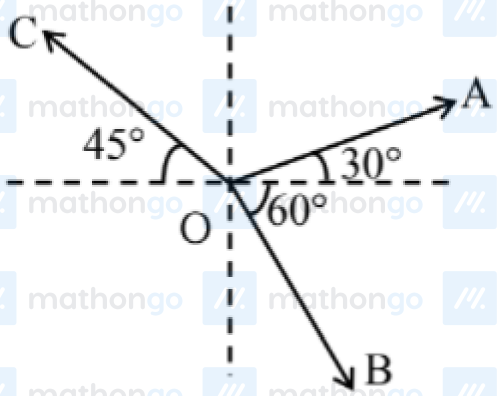
$$\tan\alpha = \frac{B\sin\theta}{A - B\cos\theta}$$

$$= \frac{B\sqrt{\frac{3}{2}}}{A - B \times \frac{1}{2}2}$$

$$\tan\alpha = \frac{\sqrt{3}B}{2A - B}$$

Ans (3)

Q8 (1)



Let magnitude be equal to λ

$$\overrightarrow{OA} = \lambda\left[\cos 30^\circ \hat{i} + \sin 30 \hat{j}\right] = \lambda\left[\frac{\sqrt{3}}{2}\hat{i} + \frac{1}{2}\hat{j}\right]$$
$$\overrightarrow{OB} = \lambda\left[\cos 60^\circ \hat{i} - \sin 60 \hat{j}\right] = \lambda\left[\frac{1}{2}\hat{i} - \frac{\sqrt{3}}{2}\hat{j}\right]$$
$$\overrightarrow{OC} = \lambda\left[\cos 45^\circ(-\hat{i}) + \sin 45 \hat{j}\right] = \lambda\left[-\frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{j}\right]$$
$$\therefore \overrightarrow{OA} + \overrightarrow{OB} - \overrightarrow{OC}$$
$$= \lambda\left[\left(\frac{\sqrt{3}+1}{2} + \frac{1}{\sqrt{2}}\right)\hat{i} + \left(\frac{1}{2} - \frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}}\right)\hat{j}\right]$$

$\therefore$ Angle with x-axis

$$\tan^{-1}\left[\frac{\frac{1}{2}-\frac{\sqrt{3}}{2}-\frac{1}{\sqrt{2}}}{\frac{\sqrt{3}}{2}+\frac{1}{2}+\frac{1}{\sqrt{2}}}\right] = \tan^{-1}\left[\frac{\sqrt{2}-\sqrt{6}-2}{\sqrt{6}+\sqrt{2}+2}\right]$$
$$= \tan^{-1}\left[\frac{1-\sqrt{3}-\sqrt{2}}{\sqrt{3}+1+\sqrt{2}}\right]$$

Hence option (1)